Introductory activity

Activity 1 · A function that turns sums into products 35 min. For every real number θ, let $f(\theta) = \cos\theta + i\sin\theta$.

1. Compute $f(0)$, $f\left(\frac{\pi}{2}\right)$, $f(\pi)$. What is the modulus of $f(\theta)$? Where does the image of $f(\theta)$ lie?
2. Write $f\left(\frac{\pi}{3}\right)$ and $f\left(\frac{\pi}{6}\right)$ in algebraic form, then compute the product $f\left(\frac{\pi}{3}\right) f\left(\frac{\pi}{6}\right)$. Compare it with $f\left(\frac{\pi}{2}\right)$.
3. Compute likewise $f\left(\frac{\pi}{4}\right)^2$ and compare it with $f\left(\frac{\pi}{2}\right)$.
4. With a calculator or a spreadsheet, compare $f(a)f(b)$ and $f(a+b)$ for three pairs (a,b) chosen at random.
5. Which known function satisfies a relation of the same kind, $g(a)g(b) = g(a+b)$ and $g(0) = 1$?
6. Expand $f(a)f(b)$. Which trigonometric formulas would one need to know to prove the conjecture?